

consumes less power. The first DDR2 modules to hit the market were faster than the fastest DDR modules available now. DDR2 is not without its drawbacks, though, at least initially: it's not backward-compatible with DDR, and motherboards need to be specifically designed to support the standard. This is not a small problem; not many motherboard manufacturers have shown interest in the new standard, because of the relatively large initial price difference between DDR and DDR2. However, as an indicator of the importance of the emergence of the standard, AMD will build support for DDR2 into the Opteron's on-board memory controller, later this year.

GDDR3

GDDR3 is RAM optimised for graphics processing. It is the fastest memory available today. Speed and power are the important factors when it comes to graphics designs; GDDR3, therefore, delivers four data bits every two cycles, and consumes half the power, of graphics DDR2. The first products you can expect to see using GDDR3 chips will be high-end graphics cards.

RLDRAM

Reduced Latency DRAM (RLDRAM) features a new design that minimises the time between the beginning of the access cycle and the first appearance of the data. RLDRAM is thus all about performance—it combines fast access due to low latency with high density and high bandwidth. The technology optimises performance

Futuristic Storage Material

Researchers at Princeton University and Hewlett-Packard Laboratories have constructed a low-cost storage device that combines the properties of solid-state silicon memory devices and plastic storage devices. It consists of thin-film silicon, covered with a conductive polymer coating that's been dubbed Pedot. This combination can potentially store one hundred megabits per square centimetre. Pedot is conductive only at low voltages, and permanently loses its conductivity when exposed to higher voltages. As a result, squares of conducting and non-conducting Pedot can be ones and zeroes. A memory card constructed using this method would have no moving parts, and would be like ROM—writable once, and readable many times. According to the researchers at Princeton and HP, the memory cards could be commercially viable in five years.

Researchers now say that hybrid

organic-semiconductor memory could become viable in as little as two years. One possibility is to use layers of molecules as bits, with the challenge being to find molecules that can withstand high temperatures and other harsh conditions typical of the silicon manufacturing process, as well as the process of writing and re-writing data. Researchers from the University of California have shown that a type of 'porphyrin' molecule holds up under temperatures as high as 400 degrees Celsius, even after being written to and read from trillions of times. The molecule can be used to store charge, so charged and uncharged molecules can be ones and zeroes. It has also been shown that one memory element—a group of molecules—can store two or three bits, instead of one. As a result, current 0.13 micron memory chips would be able to store as many bits as chips fabricated using the 0.09 or 0.07 micron processes of the future.

for cache and networking applications. Although lower latency would certainly mean better performance for personal computers, RLDRAM is being targeted mostly at the networking equipment segment, such as routers and switches.

FCRAM

Fast Cycle RAM (FCRAM) was developed by Fujitsu in 1999. It's radical in its

approach; it changes the core of the DRAM architecture, whereas other technologies have obtained speed improvements by enhancing the components that accessed the memory core. Changes to the DRAM core include core segmentation and pipelined operation. This is the first time the core of the DRAM technology has been changed, in more than twenty years. And the changes make for amazing possibilities: row and column information can be sent at the same time in FCRAM, and, because of pipelined operation, a secondary command can be issued without waiting for the primary command to be fully executed. The interface to FCRAM can be both SRAM-like, which will be used in mobile phone applications, and Single Data Rate (SDR) or DDR, which can be used in PC systems.

IRAM

One approach to reduce the problem of processor-memory communication delay is Intelligent RAM (IRAM)—integrating processor and memory on a single die. Of course, for the solution to be complete, memory latency needs to be reduced, too. Currently, IRAM products are not commercially available; however,

